



AMOGH H

Embedded Systems & Field
Engineering | ECE Graduate | BMC
Qualified



9108555712



amogh.harsha2501@gmail.com



Mysore Karnataka | Open to
relocation and extensive field
travel



[Portfolio Website](#)

EDUCATION

B.E. in Electronics and
Communication
Engineering

**National Institute of
Engineering (NIE), Mysuru**

2022 - 2026 (June)

FIELD & OUTDOOR TRAINING

Basic Mountaineering
Course — 2026

**Atal Bihari Vajpayee
Institute of Mountaineering
and Allied Sports - ABVIMAS**

STRENGTHS

- Hands-on hardware and firmware troubleshooting
- Sensor integration, testing and system debugging
- Field deployment and remote-site readiness
- Hardware-software system integration
- Rapid prototyping and problem solving

About Me

Electronics and Communication Engineering graduate with hands-on experience in embedded systems, IoT, sensor integration, firmware development and hardware troubleshooting. Experienced in developing and testing real-world systems using ESP32, LPC1768, BLE, Wi-Fi and environmental sensors, with exposure to PCB debugging, actuator control and hardware-software integration.

Completed the Basic Mountaineering Course (BMC), with experience operating in physically demanding outdoor and remote environments. Comfortable with extensive travel, field deployments, equipment setup, on-site troubleshooting and working alongside multidisciplinary teams. Interested in defense, UAV, autonomous systems, surveillance and mission-critical field engineering roles.

Embedded & Mobile Systems Intern — Genius Industrial Services

Sep 2025 – 2026 | Mysuru, Karnataka | Hybrid

- Developed and tested an ESP32-based connected control system integrating temperature sensing, water-level sensing, relays, BLE and Wi-Fi communication.
- Built embedded firmware for real-time sensing, equipment control and communication with a mobile application.
- Performed sensor integration, firmware debugging and hardware-software troubleshooting during prototype development and testing.
- Worked across electronics, firmware and mobile layers to diagnose system issues and improve operational reliability.
- Collaborated with hardware and product teams during integration and pilot-stage development.

Technologies: ESP32, Embedded C/C++, BLE GATT, Wi-Fi, Sensors, Relays, Flutter

Embedded Systems Intern — 3Zero

Jul 2024 – Jan 2025 | Mysore, Karnataka | On-Site

- Worked hands-on with sensor interfacing, actuator control, microcontrollers and embedded firmware.
- Performed PCB-level debugging and troubleshooting during prototype development.

- Comfortable working in physically demanding environments
- Cross-functional teamwork across electrical, mechanical and software systems
- Technical documentation and systematic fault diagnosis

SKILLS

- **Embedded Systems:** ESP32, LPC1768 ARM Cortex-M3, Arduino, Embedded C/C++, firmware development, sensor interfacing, actuator control, PCB debugging
- **Sensors & Interfaces:** DHT22, MQ135, DS18B20, ADC, UART, BLE GATT, Wi-Fi
- **Programming:** C/C++, Embedded C, Python, Kotlin, Dart, JavaScript
- **AI & Computer Vision:** YOLOv5/YOLOv8, OpenCV, DeepSORT, MediaPipe, PyTorch inference
- **System Development:** Hardware-software integration, prototyping, debugging, testing, fault diagnosis
- **Software:** Flutter, Jetpack Compose, Firebase, Git

- Supported integration and testing across mechanical, electrical and software subsystems.
- Assisted in taking embedded prototypes from initial development through functional system testing.
- Developed practical experience diagnosing faults in real hardware rather than simulation-only systems.

Technologies: Embedded C/C++, Microcontrollers, Sensors, Actuators, PCB Debugging

App Development Intern – Regain

Jan 2026 – Apr 2026 | Bengaluru, Karnataka | On-Site

- Contributed to development and improvement of production features across the Regain and BrainPal applications.
- Worked within an existing product codebase to implement, test and refine user-facing functionality.
- Collaborated with the product and development team to troubleshoot issues, integrate features and improve application behaviour.
- Gained hands-on experience working in an on-site product-development environment with real users and release requirements.

Technologies: Kotlin, Jetpack Compose, Firebase, APIs

PROJECTS

IoT-Based Real-Time Environmental

Monitoring System

ARM Cortex-M3 LPC1768 | DHT22 | MQ135 | UART | ADC

- Designed and developed an embedded monitoring system for real-time temperature, humidity and air-quality measurement.
- Interfaced DHT22 and MQ135 sensors with the LPC1768 microcontroller.
- Implemented 12-bit ADC acquisition for air-quality measurements and UART communication for real-time data output.
- Developed sensor timing and acquisition logic and implemented threshold-based alerts.
- Tested sensor behaviour and system response under changing environmental conditions.

AI-Powered Behaviour Detection & Surveillance System

YOLOv8 | DeepSORT | MediaPipe | PyTorch | OpenCV

- Developed a computer-vision surveillance pipeline for detecting and tracking people and identifying events such as running, fighting and loitering.
- Implemented YOLO-based detection with DeepSORT multi-object tracking and stable target IDs.
- Integrated pose-estimation and behavioural analysis for event detection.
- Designed the system around real-time monitoring and automated alert generation.
- Worked on combining detection, tracking and event-analysis components into one deployable surveillance pipeline.